



Chemical proteomics unveils that seventy flavors pearl pill ameliorates ischemic stroke by regulating oxidative phosphorylation

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ABSTRACT

Ischemic stroke has high mortality and morbidity rates and is the second leading cause of death in the world, but there is no definitive medicine. Seventy Flavors Pearl Pill (SFPP) is a classic formula in Tibetan Medicine. Clinical practice has shown the attenuation effect of SFPP on blood pressure disorders, strokes and their sequelae and other neurological symptoms, but its mechanism remains to be elucidated. In this study, we established three animal models in vivo and three cell models to evaluate the anti-hypoxia, anti-ischemia, and reperfusion injury prevention effects of SFPP. Quantitative proteomics revealed that oxidative phosphorylation (OXPHOS) is essential for SFPP's efficacy. Then, cysteine-activity based protein profiling technology, which reflects redox stress at the proteome level, was employed to illustrate that SFPP brought functional differences of critical proteins in OXPHOS. In addition, quantitative metabolomics revealed that SFPP affects whole energy metabolism with OXPHOS as the core. Finally, we performed a compositional identification of SFPP to initially explore the components of potential interventions in OXPHOS. These results provide new perspectives and tools to explore the mechanism of herbal medicine. The study suggests that OXPHOS could be a potential target for further research and intervention of ischemic stroke treatment.

1. Introduction

According to the Global Burden of Disease Study [1], stroke is the second leading cause of death worldwide, with the highest prevalence in Asia. Because of risk factors and substance abuse, the morbidity population tends to be younger [2,3]. Stroke is mainly classified as the ischemic (cerebral infarction) or hemorrhagic (cerebral hemorrhage and subarachnoid hemorrhage) according to its pathogenesis [4]. Most stroke patients suffer ischemic stroke [5]. The important etiological factor of ischemic stroke is cerebral artery occlusion, which can lead to local hypoxia damaging brain tissues [6]. Thrombolysis and stroke units

are the standards of care worldwide, but the risk of bleeding and organizational challenges impede the extensive adoption of such care in China [5]. In modern medicine, most of the treatment with antiplatelet and other drugs together with thrombolysis could reduce the risk of stroke in select patients with high-risk TIA and minor stroke [7]. There are also some neuroprotective agents used clinically, such as edaravone [8] and nimodipine [9], but still have the disadvantage of affecting multiple cell types throughout the body and associating with abnormal results in prothrombin time and platelet count [10]. Ischemic stroke has a complex pathophysiology which includes perturbations in the brain energy metabolism processes [11,12]. Brain blood flow is interrupted

Abbreviations: SFPP, Seventy Flavors Pearl Pill; OXPHOS, Oxidative phosphorylation; ABPP, Activity-Based Protein Profiling; GBD, Global Burden of Disease; MCAO, Middle cerebral artery occlusion; UCAL, Unilateral carotid artery ligation; CMC-Na, Sodium carboxymethyl cellulose; OD, Optical density; rdTOP, Reductive Dimethyl Labeling with Tandem Orthogonal Proteolysis; KEGG, Kyoto Encyclopedia of Genes and Genomes; GO, Gene Ontology; SFPPCS, Seventy Flavors Pearl Pill containing serum; IA probe, iodoacetamide-alkynyl probe.

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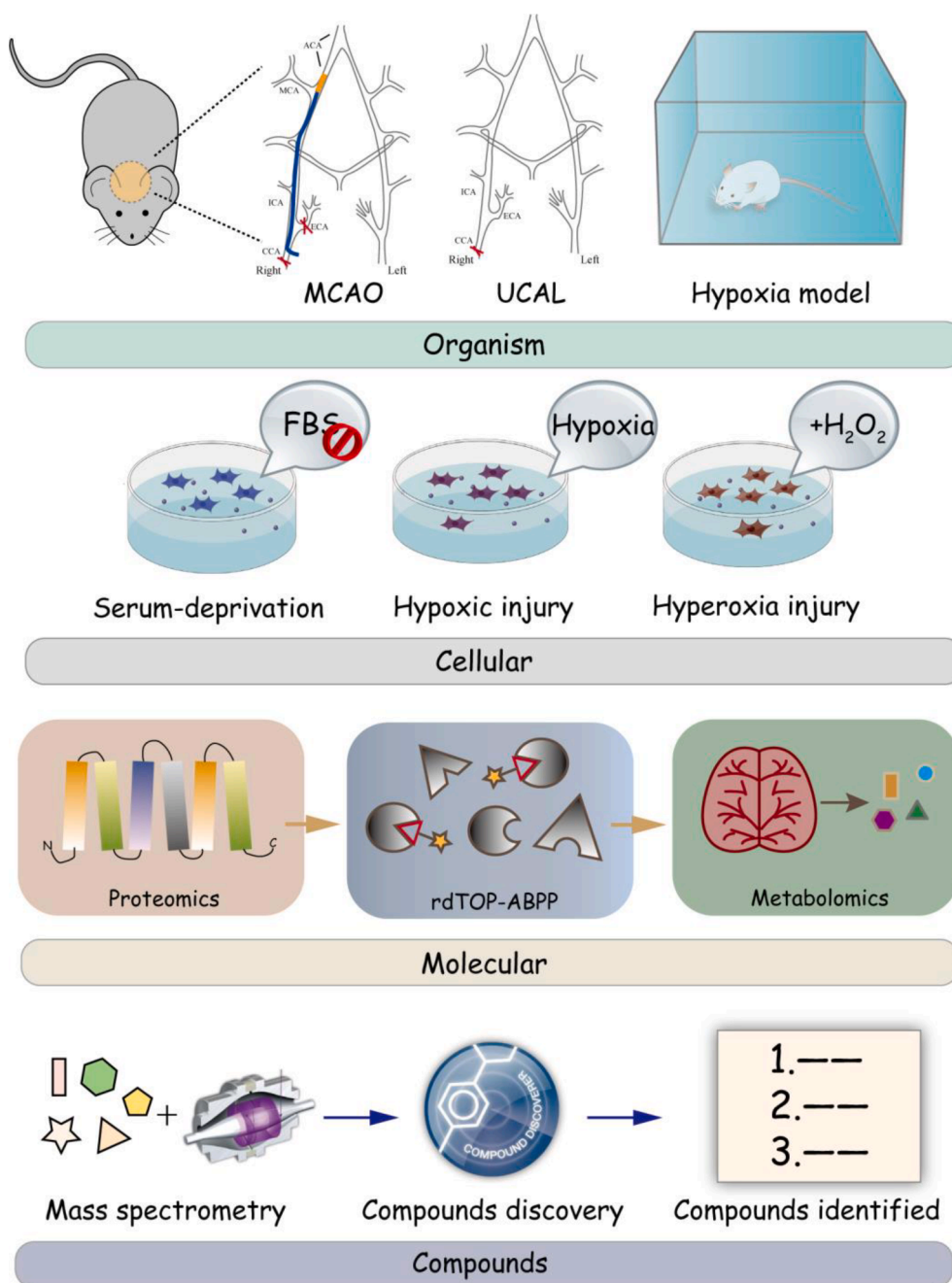


Fig. 1. The schematic of the research strategy for SFPP's anti-ischemic stroke mechanism. SFPP, Seventy Flavors Pearl Pill.

during acute ischemic stroke, which disrupts oxygen and glucose delivery, leading to mitochondrial oxidative phosphorylation dysfunction and cellular bioenergetic stress [13]. Regulation of glycolysis and oxidative phosphorylation could ameliorate brain injury after ischemic stroke [12,14]. The in-depth research of mitochondrial energy metabolism would provide further insights on the pathophysiology and facilitate the development of therapeutic interventions.

Tibetans use a unique therapeutic strategy for stroke diseases due to their geography and culture [15–17]. Seventy Flavors Pearl Pill (SFPP) is a classic formula for the treatment of central nervous system diseases [18]. In clinical practice, SFPP can effectively ameliorate the clinical symptoms of ischemic stroke [19,20], promote neurological function recovery [21], contribute to motion function and improve quality of life [22]. SFPP also exerts a significant protective effect on the blood–brain barrier in rats with cerebral ischemia reperfusion injury [23].

Today, scholars have preliminarily explored the pharmacodynamic mechanism of SFPP in brain-injury-related diseases [23–25], but the literature has been mostly clinical case reports, and modern pharmacological studies are less reported. The detailed pharmacodynamic pathways and molecular mechanisms of SFPP in protecting against ischemic stroke injury remain unclear. In addition, multiple injuries may be encountered in the pathogenesis of ischemic stroke, such as tissue damage induced by reduced blood flow, cell death caused by tissue hypoxia, and peroxide damage caused by reperfusion. Treatment strategies for ischemic stroke have to be explored from multiple perspectives.

In this work, a series of animal models and cell models were established to evaluate SFPP's effects on ischemic stroke from multiple in vivo and in vitro perspectives. Then, quantitative proteomics was used to reveal that oxidative phosphorylation (OXPHOS) is a critical pathway

for SFPP. To zoom-in the functional differences of critical proteins in OXPHOS, cysteine-activity-based protein profiling was performed. Finally, quantitative metabolomics were employed to verify that SFPP could regulate the OXPHOS to affect energy metabolism, and that the neuroprotective effect could be abolished by OXPHOS inhibitors. In addition, we also identified the migration components of SFPP and preliminarily explored the mechanism of the active ingredient. The schematic of the research strategy is presented in Fig. 1.

2. Materials and methods

2.1. Materials and reagents

In order to ensure the consistency of efficacy and eliminate variations caused by source and preparation differences, we selected standardized production of the Seventy Flavors Pearl Pill (SFPP) for this research. SFPP (Z54020062) was obtained from Tibet Ganlu Tibetan Medicine Co., Ltd. (Tibetan, China). The quality inspection of SFPP complied with the revised guidelines published on 31 December 2020 (Chinese Pharmacopoeia). The ingredients of SFPP included *Margarita* (from *Pteria martensii*), *Santali Albi Lignum* (from *Santalum album* L.), *Glycyrrhizae Radix Et Rhizoma* (from *Glycyrrhiza uralensis* Fisch.), *Bambusae Concretio Silicea* (from *Bambusa textilis* McClure), *Croci Stigma* (from *Crocus sativus* L.), *vitro-cultured Bovis Calculus* (from *Bos taurus domesticus* Gmelin), *artificial Moschus* (from *Moschus berezovskii* Flerov), *Corallii Rubra* (from *Corallium japonicum* Kishinouye), *Agate*, *DZI* (Agate-like crystals or sedimentary rocks), *Zuotai* (Multi-component mineral formulations) and other medicinal flavours. The validity of the plant names was confirmed by referring to the database at <https://www.worldfloraonline.org/> (as of 1 September 2023). Some mineral medicines lacked Latin names. Regrettably, due to the Tibetan prescription nature of SFPP and its privacy statement, we are unable to disclose all the contents, proportions, and preparation methods of the herbal components in this article.

MCAO sutures were acquired from Xinong (Beijing, China). Ethanol (j03459) was obtained from Fuyu (Tianjin, China). Human Neuroblastoma Cells (SH-SY5Y) were procured from the China Center for Type Culture Collection (Wuhan, China). Fetal bovine serum (11011-8611, FBS) was purchased from Tianhang (Huzhou, China). Minimum Essential Medium (L550KJ, MEM) and Dulbecco's Modified Eagle Medium/Nutrient Mixture F-12 (L310KJ, F12) were purchased from BasalMedia (Shanghai, China). FCCP (S8276), CCCP (S6494), IACS (S8731) and Oligomycin A (S1478) were purchased from Selleck (Shanghai, China). DADPS Biotin Azide (1330) was purchased from Click Chemistry Tools (AZ, United States). SDS (1105P031) was purchased from Solarbio (Beijing, China). Rhodamine-N₃ (X-CL-1123) was purchased from Xinyanbomei (Xi'an, China). CCK8 kit (M4839) was purchased from Abmole (TX, United States). Triton X-100 (T8787), Triethylammonium bicarbonate buffer (18597, TEAB), Urea (U5378), DL-Dithiothreitol (43815, DTT), Iodoacetamide (I1149, IAA), CH₂O (F1635), CD₂O (492620), ¹³CD₂O(596388), NaBH₃CN (42077), NaBD₃CN (190020), ammonium hydroxide solution (221228), formic acid (F0507), Tris(2-carboxyethyl)phosphine (75259, TCEP), Tris(benzyltriazolylmethyl)amine (91077, TBTA), CuSO₄ (PHR1477) were purchased from Merck (Darmstadt, Germany). BCA protein assay kit (23225), Methanol (A456-4), Acetonitrile (A955-4), Isopropanol (A461-4), Pierce™ C18 Tips (87782), Streptavidin Agarose (20353, beads) were purchased from Thermo Fisher Scientific (MA, United States). Isoflurane (R510-22-10) were obtained from RWD (Shenzhen, China). Trypsin (V5280) was purchased from Promega (WI, United States). All metabolite standards were purchased from Sigma Merck (Darmstadt, Germany), Steraloids (Beijing, China), and TRC Chemicals (YTO, Canada). Superoxide Dismutase (SOD) assay kit (A001-3-2), Malondialdehyde (MDA) assay kit (A003-1-2), Glutathione Peroxidase (GSH-Px) assay kit (A005-1-2), Lactate dehydrogenase (LDH) assay kit (A020-2-2) were purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China).

2.2. Animal experiment

Male Wistar rats (average weight 250–280 g) and KM mice (average weight 18–20 g) were purchased from the Laboratory Animal Center of Lanzhou Veterinary Research Institute, Chinese Academy of Agricultural Sciences, with the Animal Production License of SCXK (Gan) 2020-0002, raised in the Animal Experiment Center of Lanzhou University and freed to supplement food and water. The animals were fed for 7 days to acclimatize before modelling. The Animal Ethics Committee of Lanzhou University approved daily animal care and experimental procedures (kindly see the supplementary document for the original certificate).

2.3. Establishment of middle cerebral artery occlusion (MCAO) model

In this research, the rats were under anesthesia with isoflurane. The MCAO strategy was used to establish an ischemia–reperfusion (I/R) injury model, as reported [26]. First, a median incision was made above the joint of the shoulder swelling to expose the cervical muscle group. The right common carotid artery (CCA), external carotid artery (ECA) and internal carotid artery (ICA) were exposed by blunt dissection of the muscle gap. The proximal end of the CCA and the distal end of the ECA were ligated and incised 0.5 cm from the intersection of the ECA and ICA, and suture (0.3 mm) was inserted into the incision to a depth of 18.0 ± 2.0 mm to induce cerebral ischemia by MCAO. Following a period of 2 h of ischemia, the suture was gently removed by stretching it out to a distance of 5 ± 2.0 mm. This action aimed to reinstate the blood flow and enable reperfusion for a duration of 24 h. Moreover, throughout the entire reperfusion period, body temperature carefully maintained at 37 °C. One hour before modeling, SFPP group was given at 40, 80 and 160 mg·kg⁻¹·d⁻¹ intragastrically as low dose, middle dose and high dose. Nimodipine was used as active drug in accordance with a previous study [9]. Active control (AC) group was given at 12 mg·kg⁻¹·d⁻¹ intragastrically according to the dose used for ischemic stroke patients in the clinic (120 mg per day), except for the administration of pure water. Neurological function after 24 h of MCAO reperfusion in rats was assessed by the Longa score method [27].

2.4. Establishment of unilateral carotid artery ligation (UCAL) model

After anesthesia, the animals were fixed on the surgical table and disinfected with iodophor. The median incision above the line of shoulder swelling was taken to expose the cervical muscles. The muscle space was bluntly separated to expose the right CCA and vagus nerve, and then the CCA was ligated with a 4-0 surgical wire. With the exception of non-ligation, the sham group was similar to the model group. Drugs were administered for seven days before and after modeling, with a total of 14 days. SFPP group was given at 80 mg·kg⁻¹·d⁻¹ intragastrically based on clinically effective dose, except for the administration of pure water.

2.5. Grid walking test

The grid walk test was used to evaluate sensory coordination [28]. Wistar rats were placed on a 4 cm × 4 cm grid for 3 min and the procedure was filmed. The number of error steps (stepping on or falling into the space) was recorded. The rat footstep error rate was calculated according to the number of error steps / total steps × 100 %.

2.6. Preparation of drug-containing serum

SFPP group was given at 80 mg·kg⁻¹·d⁻¹ intragastrically for 7 days. Blood samples were taken from the abdominal aorta at 1, 2, 4 and 8 h after administration and centrifuged at 4 °C, 12,000 rpm for 10 min. The serum was denatured by adding 4 times the volume of anhydrous ethanol and the supernatant was obtained by centrifugation (4 °C,

15000 rpm, 10 min). The evaporation of the supernatant was undertaken at 45 °C and 16500 rpm by vacuum rotary evaporator for a duration of 300 min. The control group was operated in the same way as the SFPP group, except for the administration of pure water.

2.7. Establishment of hypoxia model

Place the mice in a closed conical flask and record the survival time of each mouse. SFPP group was given at 144 $\mu\text{g}\cdot\text{kg}^{-1}\cdot\text{d}^{-1}$ intragastrically for 10 days of pre-administration. The control group was given pure water.

2.8. Cell culture and modeling

SH-SY5Y cells were cultured in culture medium (MEM: F12 medium, v/v, 1:1) with FBS and PS at 5 % CO₂ and 37 °C. SH-SY5Y cells were inoculated into 96-well plates at a density of 1×10^4 cells per well. The initial culture medium was changed to the MEM + F12 medium (0 % FBS) for 48 h as an ischemic cell model. Cells were administered with 30 μM hydrogen peroxide for 12 h and replaced with MEM + F12 medium for 48 h as a model of peroxisomal damage cells. The original medium was replaced with MEM + F12 medium (0 % FBS and 1 % PS) and cultured at 37 °C with 1 % O₂, 5 % CO₂ for 48 h as hypoxic cell model. The 1–2–4–8 h drug-containing serum was dissolved and administered according to the volume ratio of 1:1:1 before cellular experiments.

2.9. CCK8 assay

SH-SY5Y cells were inoculated into 96-well plates at 1×10^4 per well. After the cells were treated with SFPPCS, FCCP, CCCP, IACS or Oligomycin A, 10 μL CCK8 reagent was added for cell viability testing. Calculation of cell viability by measuring optical density OD at 450 nm.

2.10. Preparation of powder and plasma

SFPP was ground and prepared into powder for the assay. After administration, blood was collected from orbits at 30 min, 1 h, 2 h, 4 h, 6 h, 8 h, 12 h and 24 h, 1 % sodium heparin was used as anticoagulant, centrifuged at 4000 rpm for 10 min and the supernatant was extracted. The plasma was mixed, and the supernatant was obtained by SPE solid phase extraction technique and blown dry with N₂ for the assay.

2.11. HPLC-HRMS analysis

500 mg of powder was added to 1 mL 50 % methanol and dissolved by sonication for 1 h. The solution was adjusted to 500 mg/mL for analysis. Plasma powder was redissolved by adding 100 μL 50 % methanol, collected supernatant for analysis. The UPLC system (Thermo Fisher Scientific, USA) coupled with a Q Exactive Orbitrap (Thermo Fisher Scientific, USA), which had an ESI source, was used for chemical identification. Samples were separated on a C18 column (4.6 $\times$ 100 mm, 5 $\mu\text{mol/L}$, ACQUITY UPLC CSH) at the flow rate of 1.0 mL/min along with the column temperature of 35 °C. Mobile phases A was 0.1 % formic acid and mobile phases B was pure acetonitrile. The linear gradient elution was set as follows: 0–2 min (5 %–10 %B); 2–40 min (10 %–40 %B); 40–50 min (40 %–95 %B); 50–55 min (95 %–95 %B); 55–55.10 min (95 %–5 %B); 55.10–60.00 min (5 %–5 %B). The injection volume was set to 10 μL . Detection wavelengths: 205 nm, 254 nm and 310 nm. The parameters of mass spectrometric analysis: Positive mode spray voltage, 3.8 kV; Negative mode spray voltage, 3.5 kV; Stepped HCD collision energy, 35 %; Temperature, 320 °C; Full MS scan resolution, 35000; Scan range, 150–1500 Da; ddMS scan resolution, 17500; Sheath Gas flow, 35 arb; Auxiliary gas flow, 10 arb.

2.12. Protein digestion and dimethyl labeling

SH-SY5Y cells were resuspended in 200 μL 8 M urea. The soluble and insoluble fractions were then separated by centrifugation at 20,000g for 60 min. The protein then was diluted to 3 mg/mL. The protein diluent was reacted with 0.5 M DTT for 30 min at 37 °C, 600 rpm in the dark. Then 0.5 M IAA was added for 30 min at 35 °C, 600 rpm in the dark. Each sample was reacted with 0.5 M DTT for 15 min, 37 °C, 600 rpm. Next, 100 mM TEAB was added to dilute urea to 2 M and then incubated with 4 μg trypsin and 100 mM calcium chloride for 4 h at 37 °C, 600 rpm in the dark. Finally, 1 μg trypsin was added and incubated for 14 h at 37 °C, 600 rpm. Quantitative proteomic analysis was performed on samples with 3 biological replicates in each group.

After digestion, the peptides were reacted with 6 μL 4 % CH₂O and 6 μL 0.6 M NaBH₃CN at 22 °C, 800 rpm for 1 h (control group: CH₂O and NaBH₃CN; model group: ¹³CD₂O and NaBH₃CN; SFPP treated group: ¹³CD₂O and NaBD₃CN). The reaction was terminated by adding 24 μL 1 % ammonia and 12 μL formic acid. The supernatants were mixed together and desalted according to the instructions of Pierce™ C18 Tips.

2.13. Reductive dimethyl labeling with tandem orthogonal proteolysis-activity-based protein profiling (rdTOP-ABPP)

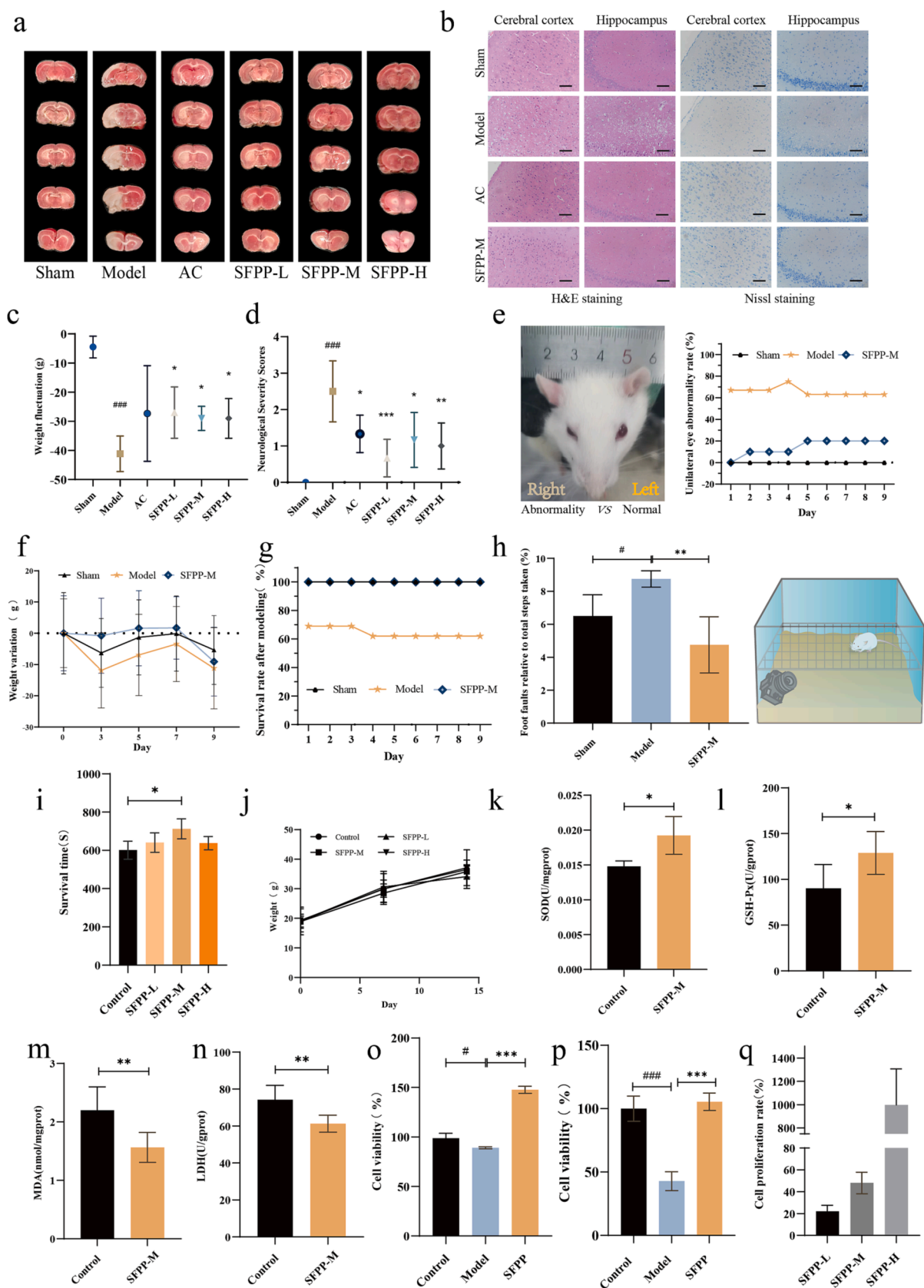
SH-SY5Y cells were lysed with 0.1 % Triton-100 by ultrasonication. Each group also had 3 biological replicates for rdTOP-ABPP. The lysate concentration was adjusted to 2 mg/mL, 20 mM IA probe was added and incubated at 25 °C, speed 800 for 1 h in the dark. A mixed reagent (TCEP:TBTA:CuSO₄ = 1:3:1) and DADPS were added for click reaction at 25 °C, speed 1000 for 1 h. After centrifugation (8500 g, 4 °C, 5 min), 300 μL pre-cooled methanol was added to the precipitate and sonicated (AMPL: 20 %), and the supernatant was discarded. 1 mL 1.2 % SDS was added to the precipitate and sonicated (AMPL: 30 %) for mixture. The supernatant was centrifuged (19,000g, 2 min) after heating in a metal bath at 90 °C for 10 min. Keep samples cooled to room temperature, 200 μL beads were added and co-incubated for 4 h at 29 °C. Discard the supernatant, 6 M urea and 25 μL of 200 mM DTT were added for the reaction at 37 °C for 30 min. Then 25 μL 400 mM IAA was added for 30 min reaction at 35 °C in the dark. The beads were incubated at 37 °C, 1200 rpm for 16 h with 2 $\mu\text{g}/\mu\text{L}$ trypsin. Dimethyl labeling was carried out in the same way as described above, in brief, 12 μL of 4 % CH₂O, CD₂O or ¹³CD₂O and 12 μL 0.6 M NaBH₃CN or NaBD₃CN were added for reaction (29 °C, speed 900, 1 h). Then, 48 μL 1 % ammonia and 24 μL formic acid were added to terminate the reaction. Following the instructions for Pierce™ C18 Tips, the supernatants of each group were mixed and desalted.

2.14. LC-MS/MS test and Data analysis

Experiments were performed on a Q-Exactive Orbitrap mass spectrometer (Thermo Fisher Scientific, LC-MS/MS) coupled to an Ultimate 3000 LC system. Briefly, the flow rate was set to 0.3 $\mu\text{L}/\text{min}$ and the distal spray voltage to 2.8 kV. MS2 scan of the top 20 ions with dynamic exclusion enabled. The mass spectrometry files were imported into pFind software for library searching and the results were analyzed by David for KEGG pathway and GO analysis. The proteomic data were deposited to the ProteomeXchange consortium (<https://proteomexchange.org>) via the iProX partner repository [29,30] with the dataset identifier PXD038535.

2.15. In-gel fluorescence analysis

After SH-SY5Y cells were lysed, the protein concentration was diluted to 2 mg/mL. IA probe was added to the lysate at 25 °C, speed 800 for 1 h. A mixed reagent (TCEP: TBTA: CuSO₄ = 1:3:1) and rhodamine-N₃ were added for reaction at 25 °C. The samples were separated by 10 % SDS-PAGE. Rhodamine fluorescence images were captured on a



(caption on next page)

Fig. 2. The pharmacodynamic results of SFPP in alleviating ischemic stroke (a) TTC staining of brain sections after 24 h of MCAO reperfusion in rats. Surviving tissues were stained red. (b) Hematoxylin and eosin (H&E) staining and nissl staining of brain sections after 24 h of MCAO reperfusion in rats. Scale bars, 100 μ m. (c) Weight fluctuation after 24 h of MCAO reperfusion in rats. (d) Neurological severity analysis after 24 h of MCAO reperfusion in rats. Data are means $\pm$ SD, $n = 6$. $^{###} P < 0.001$, compared with sham group. $^* P < 0.05$, $^{**} P < 0.01$ and $^{***} P < 0.001$, compared with model group. (e) Diagram of unilateral eye abnormality and fluctuating trend of unilateral eye abnormality after rat modeling. (f) Weight diversity curve of animal experiments. (g) Survival rate of rats in each group after UCAL. (h) Behavioral analysis of the protective effect of SFPP on UCAL model rats. Data are means $\pm$ SD, $n = 6$. $^{\#} P < 0.05$, compared with sham group. $^{**} P < 0.01$, compared with model group. (i) SFPP increased the survival time of mice under hypoxic conditions. (j) Weight diversity curve of mice. (k) SFPP increased SOD content in brain tissue of hypoxic mice. (l) SFPP increased GSH-Px content in brain tissue of hypoxic mice. (m) SFPP decreased MDA content in brain tissue of hypoxic mice. (n) SFPP decreased LDH content in brain tissue of hypoxic mice. (o) SFPPCS promoted SH-SY5Y cell proliferation. (p) SFPPCS promoted proliferation of SH-SY5Y induced by hydrogen peroxide. (q) Low, medium and high SFPPCS promoted SH-SY5Y cell proliferation under hypoxic conditions. Data are means $\pm$ SD, $n = 6$ wells from one representative of three independent experiments. $^{**} P < 0.01$ and $^{***} P < 0.001$, compared with model group. $^{\#} P < 0.05$ and $^{###} P < 0.001$, compared with control group. SFPPCS, Seventy Flavors Pearl Pill-containing serum. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Tanon imaging system.

2.16. SOD, MDA, GSH-Px and LDH analysis

SOD, MDA, GSH-Px and LDH analyses were performed based on the instructions of the kit. All mice were euthanised and brain tissue was harvested after treatment with hypoxia for 8 min. The brain tissues were prepared for 10 % tissue homogenate and centrifuged at 2000 rpm for 5 min. Then 20 μ L supernatant was mixed with the detecting reagents for SOD, MDA, GSH-Px and LDH analysis. The OD values were determined for calculation.

2.17. Quantitative metabolomics analysis

Ten mg of brain tissue were added to 20 μ L of deionized water to create a homogenate. The process of derivatization was carried out in accordance with the protocol published [31]. Quantification of all the specific metabolites in this study was conducted using the ultra-performance liquid chromatography coupled to tandem mass spectrometry (UPLC-MS/MS) system (ACQUITY UPLC-Xevo TQ-S, Waters Corp., Milford, MA, United States). UPLC-MS/MS raw data files were processed using MassLynx 4.1 software for peak integration, calibration and quantification of each metabolite. Relevant metabolic pathways were obtained by importing diverse metabolites from each group into KEGG (<https://www.kegg.jp/>) and MBRole 2.0 (<https://csbg.cnb.csic.es/mbrole2/>).

3. Results

3.1. Seventy Flavors Pearl Pill could significantly ameliorate ischemic stroke and its related syndromes

As SFPP could effectively alleviate the clinical symptoms of ischemic stroke [19,20], so we used the clinically effective dose as middle dose for animal experiment verification. To confirm the therapeutic efficacy of SFPP on ischemic stroke, we first established the classical MCAO (Middle cerebral artery occlusion) model. By TTC (Triphenyl tetrazolium chloride) staining of brain tissues, we discovered that SFPP could significantly reduce the local death of brain tissues caused by I/R (ischemia–reperfusion) injury (Fig. 2a). Macroscopically, the neurons in the cerebral cortex and hippocampus region of rats in the sham and SFPP groups were well structured, tightly arranged, with rounded nucleus, clear staining and dark blue nucleus. In the model group, the neurons were irregular in structure, with obvious nuclear consolidation, loose arrangement and a significant reduction in the number of neurons (Fig. 2b). In addition, we applied nissl staining to evaluate the status of neurons. Nissl bodies (dark blue granules) within neurons were significantly reduced after I/R injury and significantly reversed after SFPP intervention (Fig. 2b). We also observed less weight fluctuation (Fig. 2c) and lower behavioural impairments (Fig. 2d) in SFPP group after 24 h of MCAO reperfusion in rats. Arterial stenosis is also one of the causes of ischemic stroke. We simulated the process of chronic ischemia caused by

arterial stenosis through unilateral carotid artery ligation and performed drug intervention in advance for prevention and prognostic treatment after modeling. In this experiment, we observed and recorded abnormal changes in the unilateral eyes of rats. We believed that the abnormal conditions of the unilateral eyes in rats were highly related to brain ischemia caused by unilateral carotid artery ligation. The results indicated that 60–70 % of the rats in the model group had unilateral eye abnormalities, and this pathological phenomenon was significantly reduced after SFPP intervention (Fig. 2e). According to the fluctuation rate, the body weight of rats in the model group exhibited a downward trend compared with that in the sham group, whereas the body weight of rats in the SFPP group had no significant change (Fig. 2f). Meanwhile, SFPP significantly improved the survival decline caused by the UCAL (Unilateral carotid artery ligation) model (Fig. 2g). Brain injury caused by ischemia is usually reflected as central motor disorders and posture abnormalities. Referring to the published neurobehavioral experiments [28], we used a grid walking test to evaluate the severity of brain injury in rats with error rates. The results indicated that SFPP could alleviate the elevated error rate caused by the UCAL model (Fig. 2h).

The pathogenesis of ischemic stroke is caused by hypoxia and results in brain tissue damage. SFPP was effective against ischemic stroke; Therefore, we speculated whether SFPP also had a protective effect against brain damage caused by hypoxia. We established a hypoxic environment to verify that medium dose SFPP significantly improved the survival time of mice (Fig. 2i), with little effect on the body weight (Fig. 2j).

Oxidoreductases include a large class of enzymes that catalyze biological oxidation / reduction processes. It has been suggested that increasing antioxidant enzymes in neuronal cells, such as SOD and GSH-Px, can protect a variety of cells from damage during the early and late stages of brain tissue injury [32]. We examined the content of various oxidoreductases in the brain tissues of mice after hypoxia. The results indicated that SOD and GSH-Px were significantly elevated (Fig. 2k, l) and MDA and LDH were significantly reduced (Fig. 2m, n) in the brain tissues of mice in the SFPP group. This evidence suggested that SFPP could exert anti-hypoxic effects by increasing the activity of enzymes such as SOD and GSH-Px in brain tissues and thus scavenging superoxide anion radicals. The decrease in MDA and LDH indicated a significant reduction in tissue peroxidative damage, suggesting that SFPP could alleviate the damage caused by free radicals in brain tissue.

The aforementioned evidence suggested that SFPP has a pharmacological effect in alleviating ischemic stroke and related syndromes.

3.2. Seventy Flavors Pearl Pill containing serum could promote SH-SY5Y cell proliferation in various models (serum deprivation, hypoxia model and hyperoxia injury)

Ischemia observed in stroke patients could lead to damage and death of various cell types in the brain, including neurons, glial cells and blood vessels [6]. Multiple injuries may be encountered in the pathogenesis of ischemic stroke, such as tissue damage induced by reduced blood flow, cell death from tissue hypoxia, and peroxide damage caused by

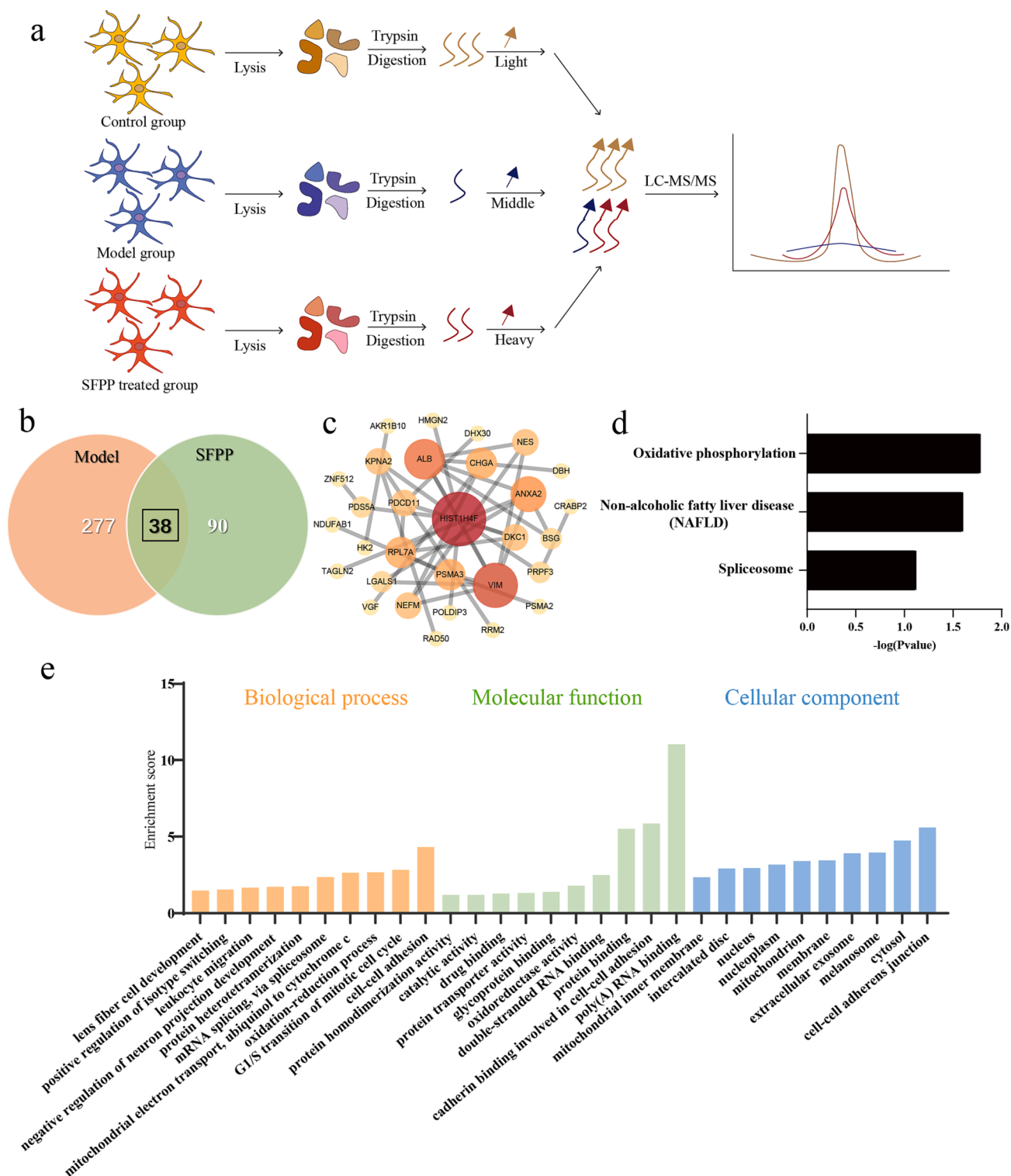


Fig. 3. Quantitative proteomics revealed critical pathways for the pharmacodynamic efficacy of SFPP (a) The experimental process of quantitative proteomics. (b) Venn diagram displaying the number of intervened proteins in the model group and SFPP-treated group. (c) Protein interaction network diagram. (d) Analysis of KEGG pathway based on quantitative proteomics. (e) Analysis of biological function based on quantitative proteomics.

reperfusion. To explore whether SFPP has a protective effect on neuronal cells (SH-SY5Y), we established “ischemic”, “hypoxia” and “hyperoxia injury” in vitro and clarified their cell phenotypes.

As a complex multidrug system, SFPP cannot be directly used for in vitro cell experiments. Therefore, we used the method of serum pharmacology [33] to perform experiments on “ischemic” nerve cells, simulating the absorption and metabolism of Tibetan medicine in vivo,

and avoiding the interference caused by in vitro experiments. Using serum pharmacology, we obtained the Seventy Flavors Pearl Pill-containing serum (SFPPCS) and used it for drug intervention on the aforementioned cells. We used CCK-8 assay to detect cell proliferation. The results of the data analysis indicated that SFPPCS has a significant effect on promoting proliferation when compared with the ischemic model (Fig. 2o). In addition, hyperoxia injury during reperfusion in the

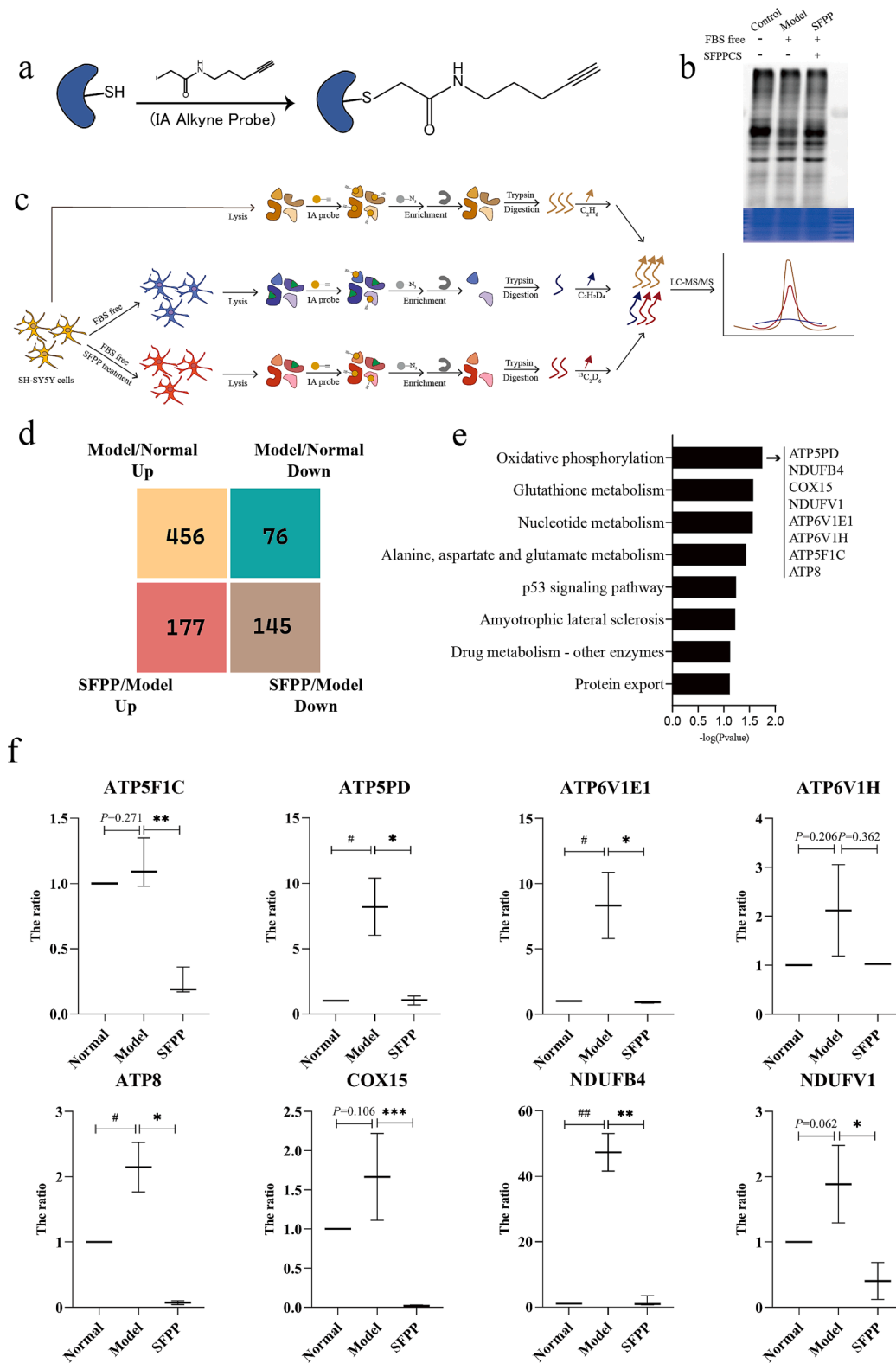


Fig. 4. Rdtop-abpp revealed that the oxphos pathway was crucial for the pharmacological efficacy of SFPP (a) The schematic diagram of IA probe binding to cysteine. (b) In-gel fluorescence analysis of active cysteines. (c) The experimental schematic of rdTOP-ABPP. (d) Analysis of intervention protein number difference. (e) Analysis of the KEGG pathway based on rdTOP-ABPP. (f) SFPP intervention reverses the disturbed key proteins of the OXPHOS pathway in model group. Data are means $\pm$ SD, $n = 3$. # $P < 0.05$ and ## $P < 0.01$ vs control group. * $P < 0.05$ and ** $P < 0.01$ vs model group. OXPHOS, oxidative phosphorylation; SFPP, Seventy Flavors Pearl Pill; and IA probe, iodoacetamide-alkynyl probe.

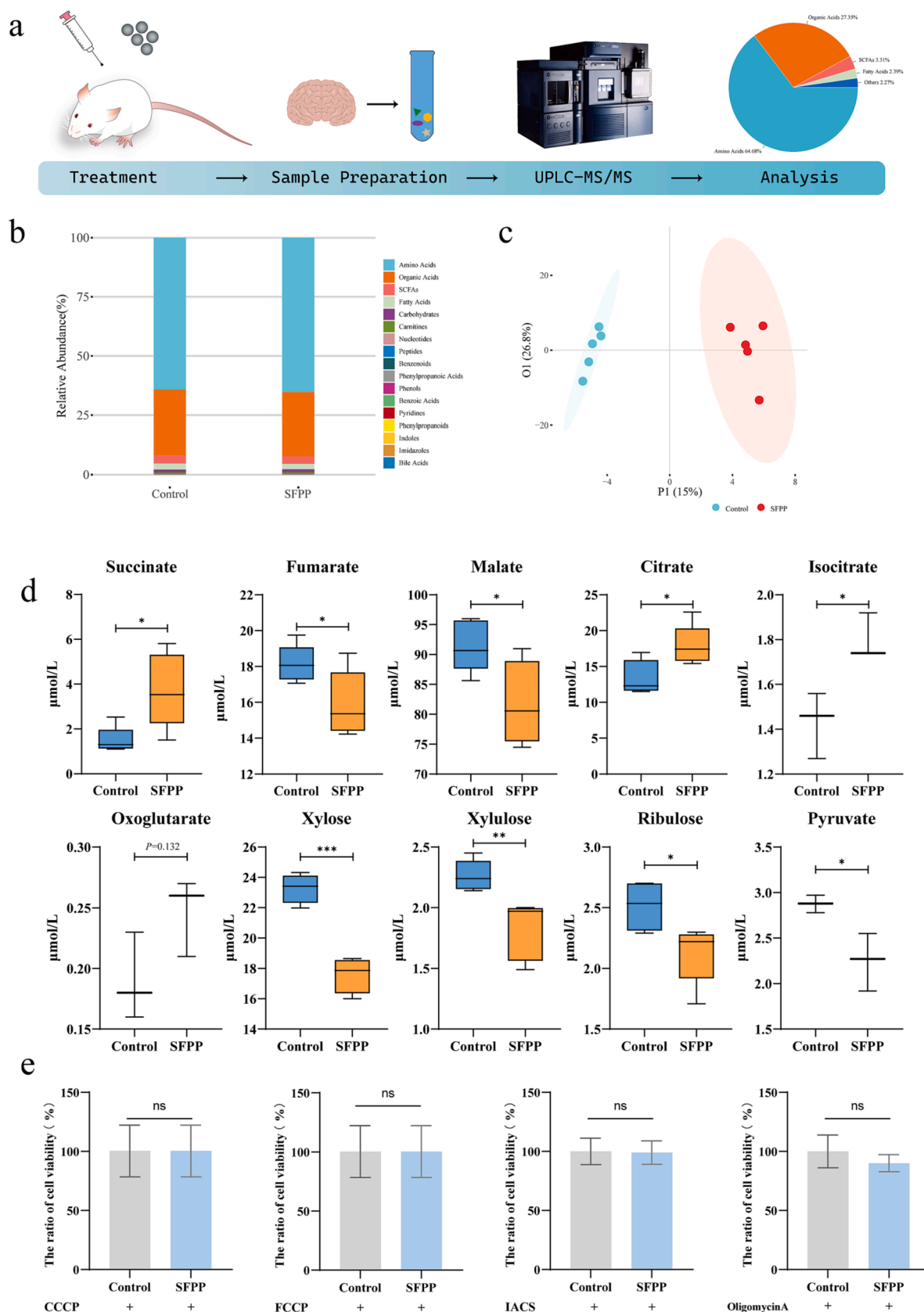


Fig. 5. SFPP could improve hypoxic injury (a) Schematic of mouse brain tissues for quantitative metabolomics study. (b) Metabolite types and relative abundance. (c) OPLS-DA analysis. (d) Biomarkers of OXPHOS and related pathways in hypoxic mouse brain tissues that underwent SFPP. Data are means $\pm$ SD, $n = 5$. ^{ns} $P > 0.05$, $*$ $P < 0.05$, $**$ $P < 0.01$, $***$ $P < 0.001$, compared with control group. (e) Viability of SH-SY5Y cells after treatment with CCCP, FCCP, IACS, Oligomycin A and SFPP. Data are means $\pm$ SD, $n = 6$ wells from one representative of 3 independent experiments. ^{ns} $P > 0.05$, compared with control group. SFPP, Seventy Flavors Pearl Pill; OXPHOS, oxidative phosphorylation; and OPLS-DA, orthogonal partial least squares discriminant analysis.

pathogenesis of ischemic stroke is an etiology of brain tissue damage. Surge of peroxides lead to tissue hyperoxia damage, so we applied direct stimulation of cells with hydrogen peroxide to model this damage phase, revealing that SFPPCS could protect neuronal cells from peroxide damage and prevent peroxide-induced cell death (Fig. 2p). The pathogenesis of ischemic stroke is caused by hypoxia and results in brain tissue damage. We demonstrated the significant promotion of SH-SY5Y cell proliferation in hypoxic conditions (1 % O₂, 5 % CO₂) (Fig. 2q).

3.3. Quantitative proteomics revealed the pharmacodynamic network and mechanism of Seventy Flavors Pearl Pill

The response of an organism to a drug or disease is a change in gene or protein networks, not a single-gene or protein change. To more comprehensively and systematically expound the biological response mechanism (such as disease mechanism and pharmacodynamic mechanism), quantitative proteomics techniques based on isotope labeling were used to comprehensively and accurately analyze the changes in protein abundance (protein quantity) in cells when drugs exert efficacy. After SH-SY5Y cells were treated with SFPPCS, the protein was extracted from the lysed cells. The stable isotope dimethyl labeling technology was used to mark the samples of each group. The samples were mixed, and the protein was analyzed with LC-MS/MS and Cimage (Fig. 3a). The results indicated that 315 proteins were regulated by the model group and 128 proteins were regulated by the SFPP-treated group, with a total of 38 proteins (Fig. 3b). Protein interaction network analysis was performed for critical proteins (Fig. 3c). KEGG pathway enrichment and GO analyses were performed using bioinformatics methods. The results indicated that the OXPHOS pathway might be an essential pathway for the efficacy of SFPP (Fig. 3d). The GO enrichment results indicated that SFPPCS could regulate biological processes such as oxidation–reduction reactions and mitochondrial electron transport, interfering with cellular components such as mitochondria and affecting the activity of redox-responsive enzymes, which are highly correlated with OXPHOS pathways (Fig. 3e). In summary, we hypothesize that the OXPHOS pathway plays a crucial role in SFPP efficacy.

3.4. rdTOP-ABPP revealed that SFPPCS could promote SH-SY5Y cell proliferation through intervention by the OXPHOS pathway

The results of quantitative proteomics indicated that OXPHOS might be a key pathway for SFPP (Fig. 3d). The oxidation reduction environment was changed when SFPP exerted pharmacological effect (Fig. 3e), and cysteine was the critical catalytic site of oxidation reductase in this process. The high nucleophilicity of cysteine thiol enables it to sensitively perceive the changes in the protein microenvironment and to further alter protein structure and function through post-translational modifications (such as nitrosylation, sulfonation, and glutathione), ultimately becoming critical for cells to cope with starvation and oxidative stress. In recent years, Benjamin F Cravatt's group has developed Activity-based Protein Profiling (ABPP) [34]. They designed an iodoacetamide-alkynyl probe (IA probe) that can covalently label active cysteines at the proteomic level, followed by the identification of marker proteins and active sites by bio macromolecular mass spectrometry, thus directly accessing structural and functional information of key proteins in a complex proteomic system. Therefore, it is favorable for us to explore the targets and pharmacodynamic network of SFPP by using the IA probe (Fig. 4a).

Fluorescence labeling experiments proved significant changes in the activity of cysteine between the SFPP and model group (Fig. 4b). Consequently, based on the published ABPP protocol [35], we established a viable experimental protocol for complex prescription research to further reveal the pharmacodynamic mechanism network of the SFPP.

After cell lysis, proteins were extracted and labeled using an IA probe. The labeled proteins were enriched using streptavidin, and the comparison between different groups was further quantitatively

analyzed using the dimethyl labeling technique (Fig. 4c). The MS results indicated that there were 532 differentially expressed proteins in the model group compared with the control group. Compared with the model group, a total of 322 differential proteins were enriched after administration of SFPP group (Fig. 4d). The KEGG pathway was enriched with intervention protein, and the primary pathway was OXPHOS (Fig. 4e). The regulated key proteins included ATP5PD, NDUFB4, COX15, NDUFV1, ATP6V1E1, ATP6V1H, ATP5F1C and ATP8. SFPP intervention reverses the disturbed key proteins of OXPHOS pathway in model group (Fig. 4f).

3.5. Quantitative metabolomics revealed that Seventy Flavors Pearl Pill affected whole energy metabolism with OXPHOS as the core

To further clarify the mechanism, we employed quantitative metabolomics to reveal the metabolites that underwent SFPP intervention. After sample pretreatment, metabolites were identified and analyzed with UPLC-MS/MS (Fig. 5a).

In the results of metabolomics research, 189 metabolites were identified and quantitatively analyzed. In both groups, amino acids and fatty acids were the main metabolites (Fig. 5b). In order to further distinguish the metabolic spectrum differences between the two groups, we used the supervised discriminant model orthogonal partial least squares discriminant analysis (OPLS-DA). Based on the results of the analysis was used, it could be observed that in the direction of the first predicted principal component, the two groups could be better distinguished (Fig. 5c).

Among the metabolite biomarkers, we noted significant changes in succinate and fumarate, two metabolites that are biomarkers reflecting OXPHOS pathways [36], and that succinate and fumarate could regulate the expression of hypoxic genes or have a close relationship with brain tissue response to hypoxia [37]. With the administration of SFPP, succinate levels were markedly increased and fumaric acid levels were reduced significantly (Fig. 5d). Succinic acid markedly enhanced OXPHOS and increased the activity of mitochondrial complex I and complex II [38]. Fumarate/succinate was decreased, suggesting that the OXPHOS pathway was regulated. In addition, succinate, fumarate, malate, citrate, isocitrate, and oxoglutarate are key metabolites involved in the tricarboxylic acid (TCA) cycle. Xylose, xylulose, and ribulose are metabolites that interfere with the pentose and glucuronide conversion pathways. Pyruvate is a biomarker that affects the glycolytic pathway. According to the metabolomics results, all the aforementioned metabolites were modulated after the administration of SFPP (Fig. 5d). TCA cycle, pentose and glucuronide conversion pathway and glycolytic pathway are closely related to the regulation of energy metabolism and affect the OXPHOS pathway. In conclusion, we suggest that SFPP could affect the energy metabolism of brain tissues via the OXPHOS pathway.

3.6. OXPHOS inhibitors blocked the pro-proliferative effect of SFPP

rdTOP-ABPP proved that the intervened proteins are mainly distributed on mitochondrial complexes I, IV and V. Metabolomics further confirmed that SFPP can play a protective role by regulating energy metabolism, thereby intervening in OXPHOS. Therefore, we selected four OXPHOS inhibitors (that is CCCP, FCCP, IACS, and Oligomycin A) to further verify our opinion.

IACS could prevent cellular respiration by inhibiting mitochondrial complex I, which means inhibiting OXPHOS by suppressing electron transfer (oxidation) in the respiratory chain. Oligomycin A could block hydrogen ion channels, which inhibits ATP synthesis and plays an inhibitory role in phosphorylation. CCCP and FCCP are uncoupling agents that act essentially by increasing the permeability of the inner mitochondrial membrane to H⁺, eliminating the transmembrane gradient of H⁺ and thus producing no ATP. Uncoupling agents affect only OXPHOS and do not interfere with substrate horizontal phosphorylation. The toxic effects of OXPHOS limited its application in

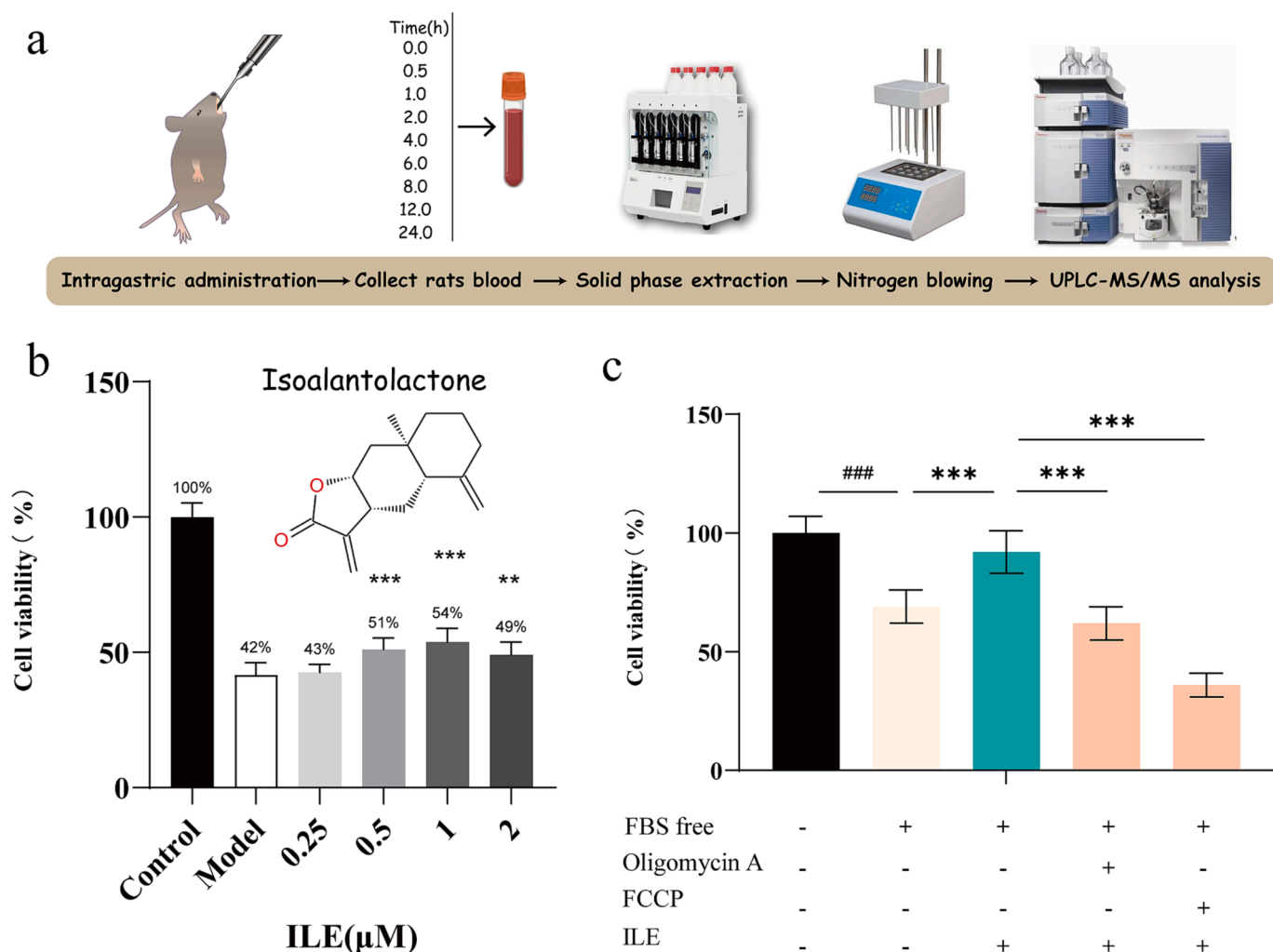


Fig. 6. Isoalantolactone promotes SH-SY5Y cell proliferation via modulating the oxidative phosphorylation pathway. (a) Schematic of rat plasma for component identification by HPLC-MS/MS. (b) ILE promoted SH-SY5Y cells proliferation. (c) Viability of SH-SY5Y cells after treatment with FCCP, Oligomycin A and ILE. Data are means $\pm$ SD, $n = 6$ wells from one representative of 3 independent experiments. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. ### $P < 0.05$ compared with control group. SFPP, Seventy Flavors Pearl Pill; OXPHOS, oxidative phosphorylation; and OPLS-DA, orthogonal partial least squares discriminant analysis.

in vivo experiments, so we performed validation experiments *in vitro*. By using a variety of OXPHOS inhibitors, we discovered that the effect of SFPP on promoting SH-SY5Y cell proliferation completely disappeared (Fig. 5e), which demonstrated that the effectiveness of SFPP depends on the OXPHOS pathway.

3.7. Isoalantolactone promotes the proliferation of SH-SY5Y cells by modulating the OXPHOS pathway

We used ultra-performance liquid chromatography-high resolution mass spectrometry (UPLC-HRMS) coupled with MS2 data to identify and characterize the major components of SFPP and plasma (Fig. 6a). The compounds were identified (Supplementary Table 1), HRMS extracted ion flow profiles were shown in Supplementary Fig. 1, components and medicinal attributes were shown in Supplementary Fig. 2a. When we attempted to use network pharmacology techniques to identify the active ingredients related to OXPHOS in the identified compounds, we did not achieve the expected results (Supplementary Fig. 2b and c). This suggested that correlative studies of components with the OXPHOS pathway are lacking and poorly supported by former researches. So, we first performed cellular pharmacodynamic evaluation experiments on the identified components and demonstrated that isoalantolactone (ILE) had a proliferative effect on SH-SY5Y cells (Fig. 6b), which was

consistent with the cellular efficacy of SFPP. In addition, we applied inhibitors of the OXPHOS pathway to perform a preliminary mechanistic investigation, showing that ILE does exert pharmacodynamic activity by interfering with the OXPHOS pathway (Fig. 6c).

4. Discussion

Ischemic stroke is one of the leading causes of disability and death worldwide and is mainly characterized by a decrease in blood flow [39]. After ischemic brain injury occurs, there are additional damage and cell death were caused by tissue hypoxia and consequent inflammatory responses [6,40,41]. With the advancement of recanalization therapy using drugs and mechanical thrombolysis, some progress has been made in helping patients recover from ischemic stroke [42]. However, at the same time, because of blood reperfusion during brain recanalization, local peroxidation surges, cause oxidative stress and secondary tissue damage [43,44]. Therefore, anti-hypoxia, anti-ischemia and reperfusion injury prevention effects are all extremely important in the treatment of ischemic stroke. In this study, we designed multiple animal models (MCAO/re, UCAL and hypoxia models) and cell models (serum deprivation, hypoxia model and hydrogen peroxide induction), simulated each stage of the disease, and explored the mechanism of SFPP's efficacy from multiple organism-cellular-molecular perspectives.

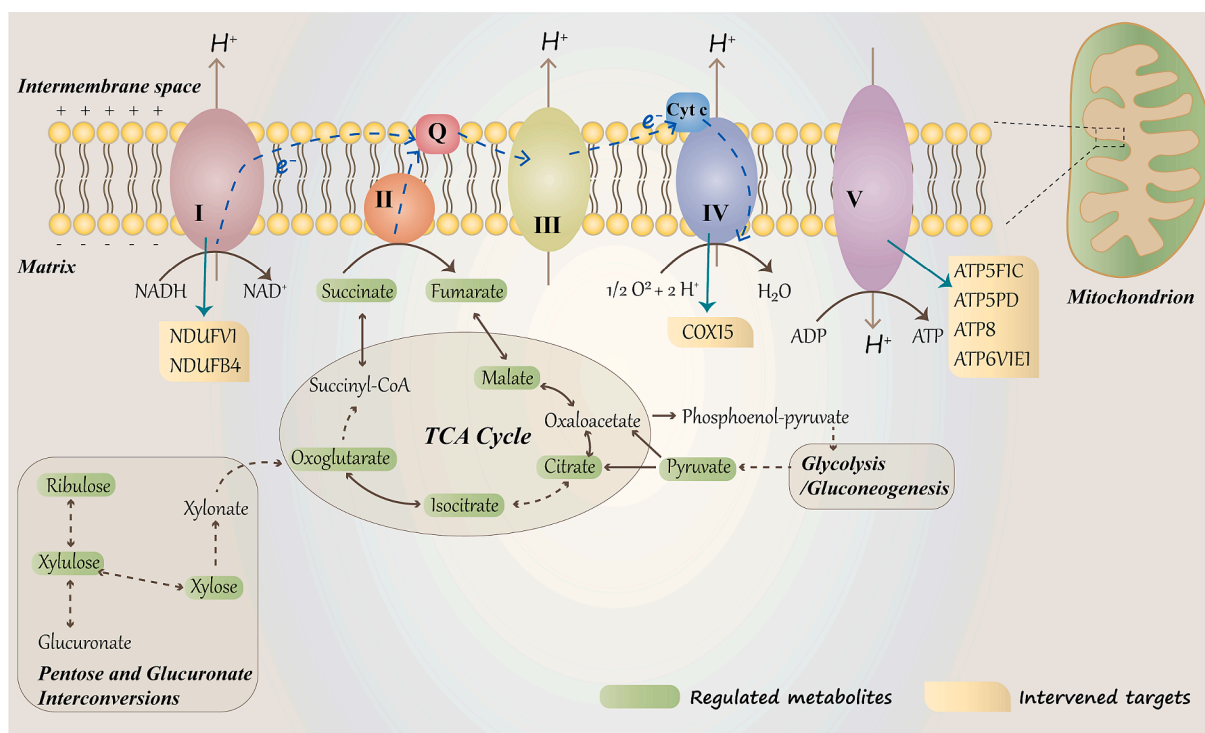


Fig. 7. Mechanism schematic of Seventy Flavors Pearl Pill in alleviating ischemic stroke via the OXPHOS pathway. OXPHOS, oxidative phosphorylation.

As a classic compound formula, Seventy-flavor Pearl Pill has a complex composition of ingredients, including botanicals, animal medicines, and minerals. In clinical trials, Seventy-flavored Pearl Pill can effectively improve the clinical symptoms of patients after stroke and improve survival quality of patients [19,20]. Meanwhile, the safety of the mineral component has attracted the attention of many researchers. It has been shown that the active blood-entering components of SFPP may have certain detoxification efficacy, which could inhibit the potential hepatic and renal toxicity of the metal components and play the role of detoxification and potentiation [45]. In addition, we also performed the evaluation of stroke and post administration hepatorenal toxicity, and the results showed that the seventy-flavored pearl pills did not enhance the hepatorenal toxicity in MCAO rats (Supplementary Fig. 3).

The response of an organism to a drug or disease is a change in gene or protein networks, not a single-gene or protein change. To more comprehensively and systematically expound the biological response mechanism, we used quantitative proteomics techniques based on isotope labeling to comprehensively and accurately analyze the changes in protein abundance (protein quantity) in cells when drugs exert efficacy. Our research revealed that SFPPCS could regulate biological processes such as oxidation–reduction reactions and mitochondrial electron transport, interfering with cellular components such as mitochondria and affecting the activity of oxidoreductase enzymes, which are highly correlated with OXPHOS pathways (Fig. 3d–e).

In recent years, it has been suggested that cerebral ischemia–reperfusion injury could be alleviated by improving intracellular energy metabolic coupling in the ischemic penumbra [46]. The main pathways involved in cellular energy metabolism include glycolysis/fermentation and OXPHOS [47], among which OXPHOS provides most of the ATP used by higher animals to support life and was responsible for regulating and maintaining metabolic homeostasis [48]. Mitochondria, the main site where OXPHOS pathways occur, are the main targets of hypoxic or ischemic injury. In addition, the lack of oxygen and nutrients leads to reduced energy dependence on mitochondria and promotes glycolysis [49]. Oxidative stress caused by tissue hypoxia is an

important part of the pathogenesis and treatment of ischemic stroke.

Cysteine, as a key amino acid to perceive protein microenvironment changes, is essential for cells to cope with starvation and oxidative stress. For the in-depth identification of active cysteine and its differential proteins, there are established IA probe and ABPP techniques. In fact, our chemical proteomics strategy based on cysteine activity is aimed at the protein perspective. Because SFPP contains complex components, including components that can directly react with cysteine residues, such as Isoalantolactone with α , β -unsaturated ketone. These can be identified as direct targets of SFPP. However, it is also possible that SFPP affects the redox environment in cells, reflecting changes in cysteine activity. These can be identified as indirect targets of SFPP. But regardless of which one, we believed those targets could be related to the efficacy of SFPP. According to the rdTOP-ABPP experiments, SFPP had intervened in the OXPHOS pathway, and the regulated key proteins included ATP5PD, NDUFB4, COX15, NDUFB1, ATP6V1E1, ATP6V1H, ATP5F1C and ATP8, which are mainly distributed on complexes I, IV and V. In summary, we suggest that modulation of the OXPHOS pathway could be a potential strategy for the treatment of ischemic stroke.

During cerebral ischemia–reperfusion injury, the tricarboxylic acid cycle, glycolytic pathways and glycogenolysis, which supply energy to the brain are disturbed [50]. It has been suggested that modulation of glucose metabolic processes can promote neuronal survival by inhibiting pyrophosphorylation [51]. Mitochondria are constantly changing to maintain normal cellular physiological functions through continuous division and fusion, providing ATP to cells through OXPHOS. Recovery of energy homeostasis and neurological function could be critical for the treatment of ischemic stroke [52].

Metabolomics experiments confirmed that SFPP could regulate many metabolites closely related to energy metabolism in the TCA cycle, pentose and glucuronide conversion pathway and glycolytic pathway (Fig. 5d) to reduce damage. In conclusion, we suggest that SFPP could affect energy metabolism by regulating OXPHOS (Fig. 7) and exerting neuroprotective effects from various stages of ischemic stroke pathogenesis.

5. Conclusions

In this research, we used multiple animal and cellular models to evaluate the pharmacological efficacy of SFPP in alleviating ischemic stroke. Through quantitative proteomics and rdTOP-ABPP technology, we hypothesize that OXPPOS is the key to the efficacy of SFPP. In this regard, through a series of quantitative metabolomics analyses and OXPPOS inhibitors, we confirmed that SFPP is dependent on OXPPOS for its efficacy in alleviating ischemic stroke. In addition, we analyzed the composition of SFPP and found ILE may have neuroprotective effects. These results provide new perspectives and tools to explore the mechanism of herbal medicine. The intervention of OXPPOS could provide a new strategy for the treatment of ischemic stroke.

6. Ethical standards

All animal experiments were complied with institutional animal care guidelines. They were conducted according to the committee-approved protocols of the ethics committee of Lanzhou University (Please refer to the supplementary document for the original certificate).

CRediT authorship contribution statement

Ruyun Ma: Writing – original draft, Methodology, Investigation, Formal analysis. **Kelsang Norbo:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation. **Yanning Zhu:** Investigation, Formal analysis, Data curation. **Chunyan Zhu:** Methodology, Investigation, Formal analysis, Data curation. **Feng Zhou:** Writing – original draft, Validation, Resources, Project administration. **Lobsang Dhondub:** Data curation, Formal analysis, Resources. **Kelsang Gyaltsen:** Investigation, Methodology, Resources. **Caisheng Wu:** Project administration, Methodology, Investigation. **Jianye Dai:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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